

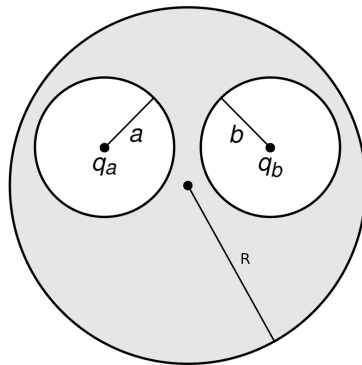
Physik II — FS 2024 — Prof. R. Wallny — Serie 5

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Exercise 1: Holes in a conducting sphere

Objective: Application of Gauss' law, electrostatic induction, properties of conductors.

Two spherical holes with radii a and b are located in a neutral, conducting sphere with radius R . A point charge (with charge q_a and q_b , respectively) is located at the center of each hole.



- Determine the surface charge densities σ_a , σ_b of the spherical holes, which inclose q_a and q_b . Furthermore, determine the surface charge density σ_R on the outside surface of the conducting sphere.
- Compute the electric field outside the conducting sphere.
- Compute the electric fields inside the holes.
- Is there a net force between the point charges q_a and q_b ? If so, compute it.
- Which of the previous answers would change if a third point charge q_c was brought from outside into the vicinity of the conductor?

Exercise 2: Surface of a conductor

Objective: Gauss' law and Stokes' law

Consider an arbitrary conductor in the context of electrostatics, where the electric field vanishes within the volume of the conductor and where $\text{rot}(\vec{E}) = 0$.

- Use Stokes' law to show that the electric field $\vec{E}$ is always perpendicular to the surface of the conductor.

- b) Show that the following relation between the surface charge density σ and the modulus of the electric field $E = |\vec{E}|$ holds at every point on the surface:

$$E = \frac{\sigma}{\epsilon_0}$$

Hint: Gauss' law may very well prove helpful.

Exercise 3: Electrostatics

Objective: Relation between electric field, potential, work, force, torque and charge distribution.

An electric field in a volume V , with boundaries $|x| \leq 1$, $|y| \leq 1$ and $|z| \leq 1$, is defined as

$$\vec{E}(x, y, z) = \begin{pmatrix} ax^2 + bz \\ cy \\ bx \end{pmatrix}, \quad (1)$$

where a , b and c are three constants (different from zero).

- Find the electric potential $\phi(x, y, z)$ that generates this field.
- How much energy is needed to move a test charge q from $P_1 = (-1, -1, -1)$ to $P_2 = (+1, +1, +1)$?
- What are the net force and torque that $\vec{E}$ exerts on the electric dipole $\vec{p}$

$$\vec{p} = p_0 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad (2)$$

which is located at $P = (x, y, z)$? At which position does $\vec{p}$ experience no torque? Is there a position where the net force on $\vec{p}$ vanishes?

Hint: The force on an electric dipole $\vec{p}$ by an electric field $\vec{E}$ is given by $\vec{F} = (\vec{p} \cdot \vec{\nabla})\vec{E}$. The torque is $\vec{M} = \vec{p} \times \vec{E}$.

- Determine the charge density $\rho(x, y, z)$ that is needed to produce $\vec{E}$? What is the total charge inside V ?
- What is the total energy of the electric field stored in V ?

Exercise 4: Laplace Operator

Objective Partial differentiation in curvilinear coordinates, Laplace equation

- Compute the two-dimensional Laplace operator $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ in polar coordinates.
Hint: $x = r \cos \phi$, $y = r \sin \phi$
- The Laplace equation for a potential Φ is $\Delta\Phi = 0$. Show that the potential of a dipole satisfies the Laplace equation in polar coordinates.
Hint: The potential of a two-dimensional dipole is $\Phi_{Dipol}(\vec{r}) = \frac{\vec{p} \cdot \vec{r}}{4\pi\epsilon_0 r^2}$. Note that this potential differs from the three-dimensional potential given in the script (3.29) by a factor r .

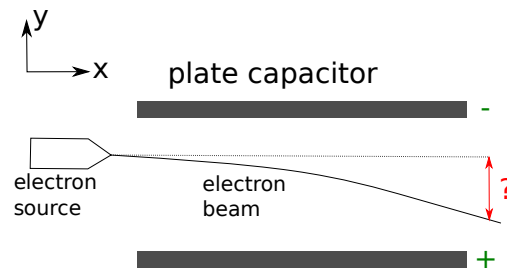
Exercise 5: Electron beam in an electric field

Objective: Moving charges in electric fields.

Consider two thin plates, both parallel to the (x, z) plane, and with a distance of d between them along the y axis (cf. figure). The bottom plate has a positive surface charge σ_+ and the top one has a negative one $\sigma_- = -\sigma_+$. The plates are infinite along the x and z axes.

- Compute the electric field that is produced by the plates in the whole space. This kind of setup is called a plate capacitor.
- Show that the electric field strength of a plate capacitor is given by $E = V/d$, where V is the potential difference of the capacitor plates and d is their distance.

Some oscillographs are based on an electron beam that is deflected by the electric field of a plate capacitor. We now consider such an electron beam that enters a plate capacitor in the x direction (see sketch). The electrons are accelerated in the electron source from standstill with a voltage of $V_0 = 1000 \text{ V}$. The length of the capacitor plates in the x direction is $s = 5 \text{ cm}$.



- Calculate the distance Δy by which the electrons are deflected in the negative y -direction on their way through the capacitor for an electric voltage of $V = 50 \text{ V}$ between the plates of the capacitor and a plate spacing of 2 cm . Assume that the field is homogenous between the plates and vanishes elsewhere.